VI. (points) In Fig. , two thin parallel plates are ideal conductors, spaced , infinite in . An EM wave between them travels normal to , along positive as a travelling wave with electric field

. is

angular frequency; is time, are unknown.

Combining these makes practical microwave transmitting antennas, replacing huge conventional parabolic dishes and greatly reducing antenna cost.

Fig.

Prove that can take only these values:

For , find

Stack equal-gap plates of unequal lengths, parallel and symmetric about an infinite line source at . Their right ends align; their lengths vary.
Fig. is the cross-section in the page. At the right end, a plane EM wave is to emerge along . Find the equation of the left boundary in the section

Fig.

Fig. . (Take ; the horizontal distance from the source to the stack's left end is